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MODBUS RELAYS DEVICE WITH ESTOP

User Manual

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2 GLOSSARY

Term	Description
Modbus RTU	A serial communication protocol used to transmit information between electronic devices over RS485. RTU stands for "Remote Terminal Unit."
Relay	An electromechanical switch controlled electronically to open or close circuits.
Failsafe Mode	A safety condition where the system defaults to a secure state (e.g. relays open) in case of failure.
EStop (Emergency Stop)	A safety mechanism to immediately halt operations in response to faults or hazards.
Recovery Mode	Mode activated to reset or reconfigure the device, typically when communication settings are lost.
Locate Mode	Mode used to visually identify a specific device within a group by flashing LEDs.
Relay Filter	A programmable delay to enforce minimum ON/OFF durations to prevent rapid toggling.
Projected State	The intended next state of a relay after the current filtered cycle completes.
Flash Memory	Non-volatile storage that retains data even after power-off; used in boot-time integrity checks.
Brown-Out Detector	Hardware that monitors power supply voltage to prevent operation during insufficient voltage levels.
Watchdog Timer	A safety timer that triggers EStop if Modbus communication fails within a preset duration.
Infeed Voltage	The supply voltage delivered to the relay module, monitored for faults.
Diagnostic Code	A numeric code reporting fault details used for troubleshooting.
RelayGuardian	Software tool used to configure and manage the device over Modbus.
Holding Register	Writable Modbus register storing device configuration parameters.
Coil Register	Modbus register used to control relays directly by writing ON/OFF commands.

3 ABBREVIATIONS

Abbreviation	Meaning
RTU	Remote Terminal Unit
RS485	Recommended Standard 485 (serial interface)
DC	Direct Current
AC	Alternating Current
LED	Light Emitting Diode
Tx	Transmit (Modbus LED indicator)
Rx	Receive (Modbus LED indicator)
RPN	Risk Priority Number
S/O/D	Severity / Occurrence / Detection (FMEA metrics)
EEPROM	Electrically Erasable Programmable Read-Only Memory
MSB/LSB	Most/Least Significant Byte
UINT32	Unsigned 32-bit Integer
MODBUS	Modular Data Bus (communication protocol)
lsb	Least Significant Bit
msb	Most Significant Bit
LSB	Least Significant Byte
MSB	Most Significant Byte

4 Introduction

This project features a MODBUS-RTU controlled relay device with emergency stop, infeed measurements, monitoring and failsafe modes. It is aimed at industrial systems such as CNC or equivalent.

This document provides a description of the operations of the device.

4.1 Device identification

The device identification is **MBR3**.

The hardware and the software are released together; the GitHub TAG references this design uniquely.

All engineering data can be downloaded from GitHub: *www.github.com/adarwoo/modbus_relay*.

4.2 FEATURES SUMMARY

This relay goes beyond a single Modbus relay and is packed with features:

1. Standard DIN Rail mountable PCB
2. 3 relays - up to 9.4A at 250V per output
   – Individual filtering for ON and OFF cycles for each relay
3. Modbus RTU v1.1v compliant
   – Coil control commands 01, 05 and 15
   – Input registers command 04
   – Holding registers commands 03, 06 and 16
   – Support for encapsulated interface transport (MEI) type 13
4. Operational integrity minded
   – Uses safety relays with forced conduits, 10.10^6 operations, reinforced isolation and position read back.
   – Galvanically isolated infeed voltage measurement with monitoring
   – Communication watchdog
5. Emergency Stop (EStop) management.
   – Safeguards relay operations
      - Relay failure
      - Infeed voltage fault (over, under, inverted, DC vs AC)
      - Communication loss
      - Application crash/corruption
   – Externally controlled by the Modbus server
      - Pulsed EStop
      - Resettable EStop with a push button
      - Terminal EStop for terminal cases
6. Operations statistics
   – Number of cycles
   – Running time
   – Fault codes
7. Diagnostics over Modbus
   – Infeed highs and lows
   – Infeed type and level
   – EStop causes and diagnostic.

5 OPERATIONS OVERVIEW

Operations are performed over a Modbus RTU communication bus.

A push button can be used on the device:

2. to reset an EStop when allowed.
3. to activate the *recovery mode*. (See Chapter 8 - Recovery mode)

5.1 SYSTEM STARTUP

The device starts with the EStop on (failsafe on power loss). The device self-checks and starts measuring the infeed. If all conditions are correct, the EStop is cleared.

This takes < 1s in normal conditions.

5.2 RELAY POLARITY AND FAILSAFE POSITION

The relay's failsafe position is 'OFF' - that is - the contact is opened. If the device loses power, the relays open the contacts at once - turning off the load. Security prevents them turning back-on in case of a crash.

The control of the relay always works with a positive polarity, so writing '1' will close the relay - and energize the load.

5.3 BASIC OPERATIONS

The relay works like any Modbus coil-based device, by writing the coil registers. Each relay can be controlled individually, turned ON, OFF. The relays can also be controlled in bulk.

See *Coil registers* chapter for details operations.

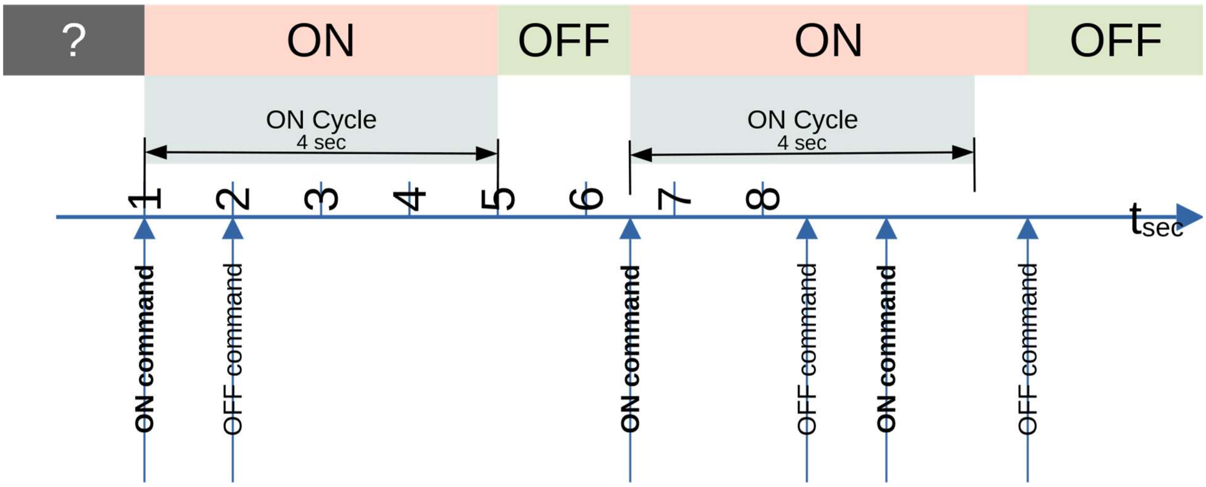
5.4 RELAY CYCLES FILTERING

Filtering can be configured to control the minimum ON and OFF relay cycles duration. An ON filter of 2 seconds means that the relay ON cycle will never be shorter than 2 seconds. The same goes for the OFF filter, which guarantees a minimum duration of the OFF cycle. Any command received during a filtered cycle will be accepted and will become the next relay state once the cycle is complete and unless overwritten by another command.

Example: a relay has an ON minimum cycle time of 4 seconds and no filter for the OFF.

4. It receives an ON command. It turns ON. The cycle starts.
5. Within 1 second, an OFF command is sent. This command becomes the next relay state, but the state of the relay stays unchanged.
6. After a further 3 seconds, the relay turns OFF since the cycle is completed

7. A new ON is sent; the relay turns ON right away since the OFF cycle is not filtered.
8. Within 2 seconds, an OFF is sent. The relay stays on.
9. Another 1 second later, an ON is sent. The relay is already ON. The next state is the current state, so nothing happens.
10. Another 2 seconds later, an OFF is sent. The relay turns OFF immediately, because the ON cycle is now more than 4 seconds, the ON cycle filter duration.



⚠ WARNING: When reading the status of a relay, the actual physical position is returned, not the projected position.

6 System Health Monitoring

This device distinguishes itself from simpler models by its ability to monitor the health of the system and to halt operations by opening an EStop relay.

The active elements being monitored and/or measured are:

1. Device integrity
2. Modbus communications
3. Infeed voltage, which is the upstream supply voltage.
4. Proper operation of the relays

6.1 Device integrity monitoring

The device monitors itself to guarantee proper execution. All memories are verified:

11. The flash memory integrity is verified at every boot.
12. The RAM is tested once on boot-up.
13. The EEprom storage has a checksum.
14. The CPU power supply is monitored by the brown-out-detector to safeguard the EEprom integrity.
15. A failsafe exists which activates the EStop if the power supply was to fail.
16. The execution is monitored; an application crash will EStop the device.

Any terminal failure leads to the EStop being activated.

6.2 Modbus communication monitoring

The device monitors the activity of the Modbus RS485 link, and through this, the health of the Modbus server. This monitoring activity can be configured to trigger a resettable EStop.

> **NOTE: The bus is considered active if frames addressed to the device are received at least once in the configured watchdog duration.**

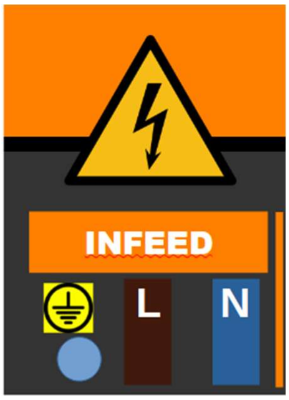
> **! IMPORTANT: For good operation, it is recommended to read the register 30009 (Status) periodically, at least twice during the configured watchdog duration.**

6.3 Infeed monitoring

The module measures the infeed voltage of the relay and the type of infeed (AC vs DC). The AC measurement is the approximate RMS value. The input signal is expected to be a 50Hz or 60Hz supply. Other frequencies will result in undefined behavior.

The last measured value, but also the lowest and highest measured values are accessible over the Modbus network. Additional Modbus registers allow resetting those values during investigations.

When applying a DC source, the negative/GND/0V must be connected to 'N' and the positive to 'P'. If this is not the case, the device will immediately go in EStop.



The in-feed measurement can be used to trigger an EStop on:

1. Incorrect type (AC vs DC) of the infeed
2. Undervoltage: The infeed voltage is lower than a configured threshold.
3. Overvoltage: The infeed voltage is higher than a configured threshold

Infeed faults can also control the relays based on individual relay configuration.

6.4 Relay health monitoring

All the relays are equipped with a position verification which is backed mechanically using a forced conduit. If a relay fault is detected:

1. The relay coil is de-energized, and the relay should become open.
2. The EStop condition is triggered.

The faulty relay must be isolated by disabling it in the configuration to clear the EStop.

The faulty relay should then be replaced.

7 LOCATE MODE

This mode allows locating a specific device in setups using more than one.

When activated, the device LEDs will alternate every second between:

1. All LEDs flash fast for 1 second.
2. All display their normal state for the next 1 second.

The locate mode is activated by a dedicated Modbus command. It is ended by either a Modbus command, or by pressing the 'EStop reset' push button.

8 RECOVERY MODE

The recovery mode is used to change the communication parameters and can be used to recover a device with unknown settings.

The recovery mode is entered by pressing the 'EStop reset' push button until the Modbus LEDs flash fast - that's around 3 seconds.

The device communication setting immediately becomes:

1. **Slave ID :** `248` ; This value is reserved in the Modbus standard, so no other devices should use it.
2. **Baud rate**: `9600`
3. **Serial setup:** `8N1` - 8bits, no parity and 1 stop bit.

You may use any Modbus master application to reconfigure the device, or use the RelayGuardian application using the following procedure:

1. Start the RelayGuardian application.
2. Make sure the communication port is properly set up in the RelayGuardian.
   The connection status should be OK
3. Issue a scan command in the RelayGuardian to make sure the RS485 adapter sends the data.
   The device Modbus LEDs should blink
4. If this is not the case, check the Modbus is plugged correctly in the PC and the comm port is the correct one.
5. If devices are identified during the scan, click the 'locate' icon one device at time to see if it is the right one.
6. Else start the recovery
   Push the relay 'EStop reset' push button for > 3s.
   The Modbus LEDs are flashing.
   In the 'RelayGuardian' application, click the 'Recovery' button.
7. Confirm proper connection.
8. In the 'RelayGuardian', click the 'Locate' button. All LEDs of relay module should be flashing fast! The module is being recovered.
9. Stop the device location mode.
10. Go to the settings tabs, check the configuration, or adjust.
11. Hit the 'Synchronize' button in the relay guardian to match the relay configuration.
    The guardian takes the relay out of recovery mode automatically.
    You're all set!

NOTE: The recovery mode keeps on-going relay operations.

TIP: The recovery mode can also be ended by pressing the 'EStop reset' push button for another 3s.

9 ESTOP MODE

The device EStop relay opens the circuit when the device is in EStop condition.

9.1 TRIGGERING OF THE ESTOP

The EStop condition is the result of:

1. Detecting an internal failure; (relay, crash, infeed, supply, ...)
2. Through external command over Modbus

9.1.1 External EStop command

The Modbus master can issue an EStop command. This external EStop can be:

1. Pulsed EStop.
   The EStop relay opens briefly stopping the system. There is reset required.
2. Resettable EStop.
   The relay remains opened until the EStop condition is cleared.
3. Terminal EStop.
   Only a power cycle or remote reset will clear the EStop condition.

9.2 CLEARING THE ESTOP

The EStop can only be cleared once the root cause has been remediated.

Clearing the EStop condition can be done:

- Automatically when the EStop is triggered remotely in pulsed mode.
- Manually, but pushing the EStop reset button.
- Remotely over the Modbus network.

In all cases, the EStop condition can be read over the Modbus network.

! IMPORTANT: When the device is in both *locate* and *EStop* mode, the first push on the EStop reset clears the *locate* mode and the second clears the EStop.

9.3 RELAYS IN ESTOP

The following EStop conditions will impact the relay position:

Condition	Impact
Application crash	All relays are open and remains open pending a manual reset of the EStop
In-feed polarity inversion	All relays are opened immediately (the filter value is bypassed). The polarity must be fixed to clear this situation
In-feed threshold	The individual relay configuration applies (register 40021). The filters are bypassed.
External EStop	No impact. The Modbus master can still control the relays
Modbus communication loss	The individual relay configuration applies (register 40022). The filters are applied.
Individual relay failure	The fault relay is opened immediately. It can no longer be controlled, and must be disabled in the configuration to clear the fault

10 LEDs

The module features LEDs to see the device operation and faults easily.
Each LED may serve multiple purposes.

For detailed LED states, see:

1. *EStop LED*
2. *Infeed LED*
3. *Modbus LEDs*
4. *Relay LEDs*

10.1 Boot

During boot, for the first 2 seconds, all LEDs are lit. This allows checking for a faulty LED. The fault LED remains on if a persistent fault condition is detected during the auto-test.

10.2 Locate function

To help locate a relays device, a Modbus *locate* command can be sent. See *Locate mode*.

In locate mode, a sequence is repeated with all LEDs flashing fast (10Hz) for one second, and display their normal value for the next.

The EStop reset push button can be pressed to end the locate cycle, or a new Modbus command can be sent.

10.3 Recovery mode

This only affects the Modbus Tx and Rx LEDs which flash synchronously at 2Hz.

10.4 EStop LED

The EStop LED state is as follows:

State	Description
Off	**Normal operations**
On	**Terminal mode** Either the device failed self-tests, or an external terminal EStop was issued. The EStop is active.
Flashes at 2Hz	**Resettable fault with EStop** It can be reset remotely or with the push button once the cause has been remediated.
ON for 2s	Shows a pulsed EStop condition received from the Modbus master

When a resettable EStop is in progress, any other LEDs flashing is an indication of the source of the EStop. See other LEDs for interpretation.

10.5 Infeed LED

The infeed LED provides visual information about the infeed voltage.

State	Description
Off	Infeed voltage is < 10V (AC+DC)
On	Infeed voltage detected, and within configured range
Flash . .	Voltage detected, but below the configured threshold
Flash _ _	Voltage detecting, but over the configured threshold
Flash at 2Hz	Along with the EStop. An infeed fault was detected.

☢ **CAUTION: The Infeed LED is for indication only.** ***You must assume voltage is always present.***

10.6 Modbus LEDs

The Modbus LEDs show activity on the Modbus RS485 network.

Both LEDs are overridden during boot, recovery and identification modes.

1. The Rx LED shows received data on the RS485 bus. The data may or may not be addressed to the device.
2. The Tx LED shows data being sent from the device (as the device replies to the master)

10.6.1 Rx LED

The following table summarizes the 'receive' LED states

State	Description
Off	No data is being received
Blinks	Data is being received on the bus. The data may or may not be addressed to the device.
Flash at 2Hz	Along with the EStop LED, indicates an EStop Modbus command was received

10.6.2 Tx LED

The following table summarizes the 'transmit' LED states

State	Description
Off	No data is being sent
Blinks	Data is being sent over the bus.
Flash at 2Hz	Along with the EStop LED, indicates a communication lost.

10.7 RELAYS LED

Each relay has a dedicated LED. The LED reports the status of the relay is corresponds to.

The LED status is as follows:

State	Description
Off	The relay is in the OFF state
On	The relay is in the ON state
Flash at 2Hz	Along with the EStop LED, indicates a faulty relay
Blink	The relay is disabled

11 MODBUS COMMUNICATIONS AND REGISTERS

The product uses *Modbus RTU over a half-duplex RS485*. It allows communication up-to 115200 baud to the device.

The Modbus registers are grouped into:

1. *Coil Registers*
2. *Input Registers*
3. *Holding Registers*
4. *Device Control Registers*

For communication settings, see *Communication Settings*.

NOTE: By convention, all relays are indexed from 0. Therefore, the device relay label 'Relay 1' is indexed at 0.

11.1 DEVICE IDENTIFICATION

The device supports Modbus 1.1b Report Slave ID function and Read Device Identification functions.

11.1.1 Report Slave ID

The function 17 is often used to scan for Modbus RTU devices on a bus.

The device implements this function.

NOTE: The Run Status indicator always reads as ON (0xFF)

The Slave ID is 'MBR3-ES_<model_name>'.

11.1.2 Encapsulated Transport Interface

The function 43 with a Modbus Encapsulated Interface Type 14 (0x2B) is supported to query device information.

The following objects are supported:

Object ID	Object Name / Description	Type	Category	Possible value
0x00	VendorName	ASCII String	Basic	ARex
0x01	ProductCode	ASCII String		MBR3-ES
0x02	MajorMinorRevision	ASCII String		2.0
0x03	VendorUrl	ASCII String	Regular	www.github.com/adarwoo
0x05	ModelName	ASCII String		1.0c
0x80	Private Object 0, number of relays	ASCII String	Extended	3

11.2 Coil registers

The relay coils can be accessed from 0 (Relay 1) up to relay 31. Coils can be written and read at will.

The device supports the following function codes:

- **01**: *Read coils* Read from 1 to 32 contiguous status of coils. Coils in the response message are packed as one per bit of a byte, 1=On and 0=Off. If the requested quantity of coils is not a multiple of 8, zeros are padded in the final byte.
- **05**: *Write single coil* Write a single output to either On (1) or Off
- **15**: *Write multiple coils* Force each coil in a sequence of coils to either On or Off

! IMPORTANT: Relays operations are not affected by the EStop Unless a relay is faulty or disabled, a command of that relay will always be executed, but could be delayed by one of the relay cycle filters

⚠ WARNING: Attempts to write individually a disabled relay with the function 05 will generate a `slave_device_failure` (4) error. Attempts to turn ON a disabled or faulty relay as part of the function 15 will also generate a slave device failure error. You can still use function **15** providing that the faulty/disabled relay is commanded to OFF.

11.2.1 Details of the function 05: Write single coil.

The following values can be used:

Value	Meaning
0x0000	This specific 16-bit value is the Modbus standard representation for "OFF".
0xFF00	This specific 16-bit value is the Modbus standard representation for "ON" when writing to a coil.
0xAA00	This value toggles the coil at the end of the cycle

11.3 Input registers

Input registers are read-only registers used to report information about the device. The registers have been grouped so they can easily be accessed. Reserved values read as 0, therefore, it is possible to read all the input registers in a single command.

Only the function **04**: *Read input registers* is supported to read registers.

11.3.1 Status & Monitoring

Modbus Address	Hex value	Access	Description	Values
30009	0x0008	R	Status	0: Device is operational 1: Device is in EStop. A reset is possible 2: Device in terminal EStop
30010-30011	0x0009-0x000A	R	Running minutes	UINT32 0 – (2^{32}-1)
30012	0x000B	R	Actual infeed voltage type	Reports the type of infeed voltage detected 0: Measured AC/DC voltages are below 10V 1: DC2: AC
30013	0x000C	R	Current infeed voltage	1/10 volts0-3000
30014	0x000D	R	Infeed highest voltage	Returns the lowest measured infeed voltage until now in 1/10th of volts. This value can be reset with the command: - *40102 - Reset measurements*
30015	0x000E	R	Infeed lowest voltage	Same as 30014, but returns the lowest
30016	0x000F	R	Device Health	Bit 0: One or more relays have faults. Check individual relay status. Bit 2: Infeed polarity inverted Bit 3: Incorrect infeed voltage type Bit 4: Infeed voltage is under Bit 5: Infeed voltage is over Bit 8: Application crash detected Bit 9: EEprom recovered Bit 10: Supply voltage failure *Bits 8 and above are cleared by a power cycle or reset command only.*
30016	0x000F	R	Last or on-going EStop root cause	Bit 0: Relay Fault Bit 1: Modbus communication lost The communication watchdog reported a lack of communication Bit 2: Infeed polarity inversion Bit 3: Unexpected Infeed voltage type An incorrect voltage type was detected. AC in vs DC expected for example. Bit 4: Infeed voltage under The voltage has gone below the configured threshold Bit 5: Infeed voltage over

				The voltage has gone over the configured threshold Bit 6: Command A Modbus command was issued to place the device in EStop Bit 8: Application crash The device is recovering from an application crash Bit 9: EEprom corrupted The EEprom memory was corrupted and was reset. Bit 10: Supply voltage failure A power failure caused the device to restart This register is not cleared when the EStop is reset, allowing for post crisis diagnostic, or checking the cause of pulsed resets.
30017	0x0010	R	Diagnostic code	Diagnostic code of the EStop condition. This register contains the last external diagnostic code supplied with the EStop
30018–30024	0x0011–0x0017	—	*Reserved*	

11.3.2 Relay Diagnostics & Statistics

The status of each relay is accessible, and well as their individual number of cycles. A cycle is defined as a change of relay state during operation (exclude power loss transitions).

The address space is structured to expand up to 16 relays for other variants of devices.

Modbus Address	Hex value	Access	Description	Values
30025	0x0018	R	Relay 1 diagnostic	0=OK 1=Faulty 2=Disabled
30026-30027	0x0019-0x001A	R	Relay 1 number of cycles	Unsigned 32-bits 0-(2^{32}-1)
30028	0x001B	R	Relay 2 diagnostic	0=OK 1=Faulty 2=Disabled
30029-30030	0x001C-0x001D	R	Relay 2 number of cycles	Unsigned 32-bits 0-(2^{32}-1)
30031	0x001E	R	Relay 3 diagnostic	0=OK 1=Faulty 2=Disabled
30032-30033	0x001F-0x0020	R	Relay 3 number of cycles	Unsigned 32-bits 0-(2^{32}-1)

⚠ **WARNING: Reading passed the last supported relay will generate an** `illegal data address` error.

TIP: For generic software, the number of available relays can be read in input registers 30004.

11.4 HOLDING REGISTERS

Holding registers hold values that can be read or written. The hold configuration details. They are grouped by functions. It is possible to read all the holding registers with 1 command as the reserved values will read as one. Writing these registers is restricted to prevent misbehavior.

⚠ **WARNING: Reserved values cannot be written and will generate an error.**

11.4.1 Supported function codes.

The following function codes are supported:

17. **03 -** *Read Holding Registers* Reads the values of one or more holding registers. Commonly used to retrieve configuration or control values stored in the device.
18. **06 -** *Write Single Register* Writes a single value to a specific holding register. Used for updating configuration or control parameters.
19. **16 -** *Write Multiple Registers* Writes a whole configuration group at once. This is only available for writing the communication settings.

⚠ **WARNING: Do not issue the function 04: Read input registers when reading holding registers as the memory addresses of both types overlaps.**

Other functions codes such as 23 (Read/Write Multiple Registers) are not supported.

11.4.2 Communication Settings

This group of registers allow configurations to configure the communication settings of the relay.

❗ IMPORTANT: These registers are write-protected. You must activate the 'recovery' mode to change them, by pressing the 'EStop Reset' button for more than 3 seconds.

NOTE: You must use function 16 - Write Multiple Registers, writing registers 40001 to 40004 at once. Once a write is done, the device comes out of recovery mode and applies the changes right away.

Modbus Address	Hex Value	Access	Description	Factory value	Values
40001	0x0000	RW1	device address	44	[1-247]
40002	0x0001	RW1	Baud rate	5	0=300 1=600 2=1200 3=2400 4=4800 5=9600 6=19200 7=38400 8=57600 9=115200
40003	0x0002	RW1	Parity	0	0=None 1=Odd 2=Even
40004	0x0003	RW1	Stop bits	1	1=1 Stop bit 2=2 stop bits
40005–40008	0x0004–0x0007	R	*Reserved*		

11.4.3 Power Infeed Configuration

Configure the expected in-feed voltage.

⚠ WARNING: Registers 40009 to 40011 must be written together using command 16

Modbus Address	Hex Value	Access	Description	Factory Value	Values
40009	0x0008	RW	Infeed voltage type	1	0=DC 1=AC
40010	0x0009	RW	Infeed lower voltage threshold	100	1/10 volts [100-3000]
40011	0x000A	RW	Infeed upper voltage threshold	3000	1/10 volts [100-3000]
40012–40016	0x000B–0x000F	R	*Reserved*	0	

11.4.4 Safety Logic Configuration

Monitored values can optionally control EStop. It is also possible to set the behavior of each relay on these optional EStop conditions.

⚠ WARNING: The registers must be written individually. Function 16 is not supported

Modbus Address	Hex Value	Access	Description	Factory Value	Values
40017	0x0010	RW	Activate EStop on undervoltage	0	0=no 1=yes
40018	0x0011	RW	Activate EStop on overvoltage	0	0=no 1=yes
40019	0x0012	RW	Activate EStop on incorrect type of supply voltage. Example: AC detected with DC configured	0	0=no 1=yes
40020	0x0013	RW	Activate EStop on number of seconds without Modbus frame received	0	0=Off 1-65535: Number of seconds
40021	0x0014	RW	Mask to open relays on electrical infeed fault, bypassing filtering. When an infeed fault is detected, the relay state is immediately opened or left unchanged based on this value. A bit at 1 in the mask indicates the relay should open on a fault detection. The lsb (bit 0) masks the relay at index 0, the msb (bit 15) masks the relay at index 15	0	0-0xFFFF Example: 0: No relays are affected 5 (binary 110): Relays at position 1 and 2 are set to open 0xFFFF: All relays are set to open
40022	0x0015	RW	Mask to open relays on Modbus communication loss fault. When a loss in communication is detected, the relay projected state[1] is opened.	0	See 40021
40023–40024	0x0016–0x0017	R	*Reserved*		

[1]The projected state is the state of a relay once the on-going filter cycle is complete.

11.4.5 Relays Configuration

This group allows configuring the individual relays.

⚠ **WARNING: The registers must be written individually. Function 16 is not supported.**

Modbus Address	Hex Value	Access	Description	Factory default	Values
40025	0x0018	RW	Relay 1 config	0=Enabled, no filter	msb bits [15-8] are for the ON filter. lsb bits [7-0] are the OFF filter. The value **0xFFFF** disables the relay. The ON and OFF filters values are given in 1/10 of seconds from 0 to 254 (0 to 25.4s) Example: 0x0132 The ON state is guaranteed to last at least 100ms, whilst the OFF state is guaranteed to last at least 50 x 1/10s = 5s after a ON. See relay operations.
40026	0x0019	RW	Relay 2 config	0=Enabled, no filter	Same as relay 1 config
40027	0x001A	RW	Relay 3 config	0=Enabled, no filter	Same as relay 1 config
40025+N[1]	0x0018+N	RW	Relay N config	1=Enabled, no filter	Same as relay 1 config

[1] N is the relay index [0... (Number of relay-1)]

⚠ **WARNING: Read or writing a non-supported relay will generate an error.**

11.5 Device control registers

These registers share the holding registers mapping address space but are write-only. This group of registers allows controlling the EStop and the device.

NOTE: These registers are write-only. The function 03: Read multiple registers is not available for this group. Function 16 - Write Multiple Registers is not available. The registers must be written individually.

Modbus Address	Hex Value	Access	Description	Values
40101	0x0064	W	Set/Reset the EStop	ESTOP_CTRL See note 1
40102	0x0065	W	Zero measurements. Allow re-measuring min and max	0xAA55
40103	0x0066	W	Identify the device. Fast flash all the LEDs to identify the device	0 = Turn off1 = Turn on
40104	0x0067	W	Reset configuration to factory default and reboot	0x178C
40105	0x0068	W	Exit recovery mode and apply comms settings	0xAA55
40106	0x0069	W	Reset the device	0xAA55

Note 1: See the format below.

11.5.1 EStop control.

The following table documents the type **ESTOP_CTRL** used to control the EStop.

Byte	Function	Values
MSB	Type of EStop	0x00: Reset the EStop if possible. Returns an error if the EStop could not be reset 0x11: Pulsed EStop. Create a 4 second EStop pulse 0x22: Resettable EStop mode. The EStop can be reset by pushing the EStop reset button 0xFF: Terminal EStop. Only a reset will clear the EStop (Register 40103)

12 ELECTRICAL AND MECHANICAL CHARACTERISTICS

For relay data, refer to the datasheet of the *SiSF2 relay* for detailed data. Note that both contacts of the relay are paired as one.

12.1 ELECTRICAL CHARACTERISTICS

Item	Min	Max
Device supply voltage	18 VDC	30 VDC
Infeed voltage	-	250V (AC/DC)
Relay contact current	-	9.4A (AC/DC)

12.2 CONTACT LIFE

Item	Value
230VAC, 1A	800k
230VAC, 9.4A	100k
24VDC, 0.5A	2M
24VDC, 9.4A	300k

12.3 TROUBLESHOOTING

12.3.1 Cannot Communicate with device.

20. If the Modbus LEDs are flickering, make sure the RelayGuardian and relay share the same settings.
21. Active the recovery mode on the device and in the RelayGuardian
22. If no communication is established, check the serial adapter communication port.
23. Check the wiring. Modbus pins cannot be swapped.

12.3.2 Lost Device ID

1. Send a valid broadcast frame to enter configuration mode.
2. Use the default Device ID (`0`) to reset the device via the reset register.

12.3.3 Relay Does Not Respond

24. Check the watchdog timeout setting to ensure it isn't triggering prematurely.

13 FMEA – Modbus Relay Board for CNC Safety

This FMEA analyzes potential failure modes of a Modbus-controlled relay board designed for CNC safety, including load control and emergency stop (E-stop) functionality. Any failure should result in the CNC stopping by letting go of the estop switch. **Note**: The EStop must be checked once at the start of operations. This step is done by the Masso CNC device. This relay EStop contact is Normally Opened.
When the CNC starts, the relay will be open. If the relay was to close **after** the Masso had booted, this could be considered an estop test. Therefore, the relay must be started well before the Masso. This is the case (100ms vs 10s).

The top events to consider are:

1. Lack of dust extraction clogs the work and leads to the cutter breaking. The operation is done in normally closed and protected condition to safeguard the operator. Mechanical damage only.
2. Lack of cooling of the spindle leads to spindle overheating. NTP sensor on the spindle should detect the overheat if wired. Else, the spindle could be damaged. Fire is unlikely.

Function	Failure Mode	Effect of Failure	Cause(s)	Detection Method(s)	S	O	D	RPN	Recommended Action(s)
Control Load Relays	Relay not switching	Load not powered leading to CNC or cutter damage	Relay failure, dry joint	Feedback circuit	8	4	3	96	Add redundant relay check, use industrial-grade relays
Control Load Relays	Relay stuck ON	Unsafe load activation	Relay contact welding	Feedback circuit	9	3	4	108	Use a forced conduit relay and read back the status of the relay
E-Stop Relay	E-stop not triggered	CNC continues in unsafe state	Logic fault, relay failure	Self-test, watchdog	10	2	3	60	Use safety-rated relay, periodic self-test
Power Monitoring	Under/over voltage undetected	Damage to CNC or relay	Sensor failure, ADC error	Voltage threshold check	9	3	4	108	Add voltage sensing, calibration check
Power Converter	Converter fails	Relay board unpowere	Component failure	Relay state monitori	7	4	2	56	Use robust converter, thermal

Function	Failure Mode	Effect of Failure	Cause(s)	Detection Method(s)	S	O	D	RPN	Recommended Action(s)
		d, failsafe triggers		ng					protection. =Power the estop relay from the mains power
Modbus Communication	Loss of communication	E-stop triggered, CNC halts	Cable fault, EMI, software crash	Timeout watchdog	6	5	2	60	Loss of comms indicates the bus is damaged or the Modbus master has crashed. EStop.
Test of the EStop switch	EStop relay opens and close cycle is considered a estop switch test	False E-stop test	Non cold reboot of the firmware	Analysis of the reboot cause	8	3	3	72	Do not allow the Modbus board to reboot

13.1 Legend

25. **S (Severity)**: 1 (low) to 10 (catastrophic)
26. **O (Occurrence)**: 1 (rare) to 10 (frequent)
27. **D (Detection)**: 1 (certain detection) to 10 (undetectable)
28. **RPN (Risk Priority Number)**: S × O × D

13.2 Summary

Focus should be placed on:

29. Improving detection of stuck or failed relays
30. Enhancing voltage monitoring redundancy
31. Ensuring robust communication and feedback diagnostics